

Levitation Power System — Design Overview

Hyperloop pod maglev drive electronics · 2 control boards + 1 power-distribution unit · pre-fabrication

1. What this is

The levitation controller outputs low-power PWM signals; this board turns those into the real, high-current drive the levitation coils need. It consists of two identical control boards (each driving two of the pod's four levitation yokes) and one power-distribution unit (PDU) that feeds both boards from a single 60 V battery.

2. How it works

Permanent-magnet biased suspension — The yokes use magnets that carry the pod's static weight at essentially zero current. The coils only need to TRIM the field up or down to hold the gap steady — so normal current is small ($< 5\text{ A}$), with brief peaks up to 30 A during disturbances.

Full H-bridge per yoke — Because the coil must be able to add AND subtract from the magnet field, each yoke is driven by a full H-bridge (four transistors) so current can flow either direction. The controller sets the current with a PWM duty cycle.

Sense and protect — Each channel measures its own coil current and has a hardware over-current trip that shuts the bridge off independently of software — a safety backstop. To reduce current it lets the coil 'freewheel' (coast down) rather than actively reversing, which keeps the bus voltage safe.

3. System architecture

The architecture follows one guiding principle: keep the high-current power path simple and centralized, and keep the control distributed. A single 60 V battery feeds a dedicated power-distribution unit (PDU), which fans the bus out in parallel to both control boards.

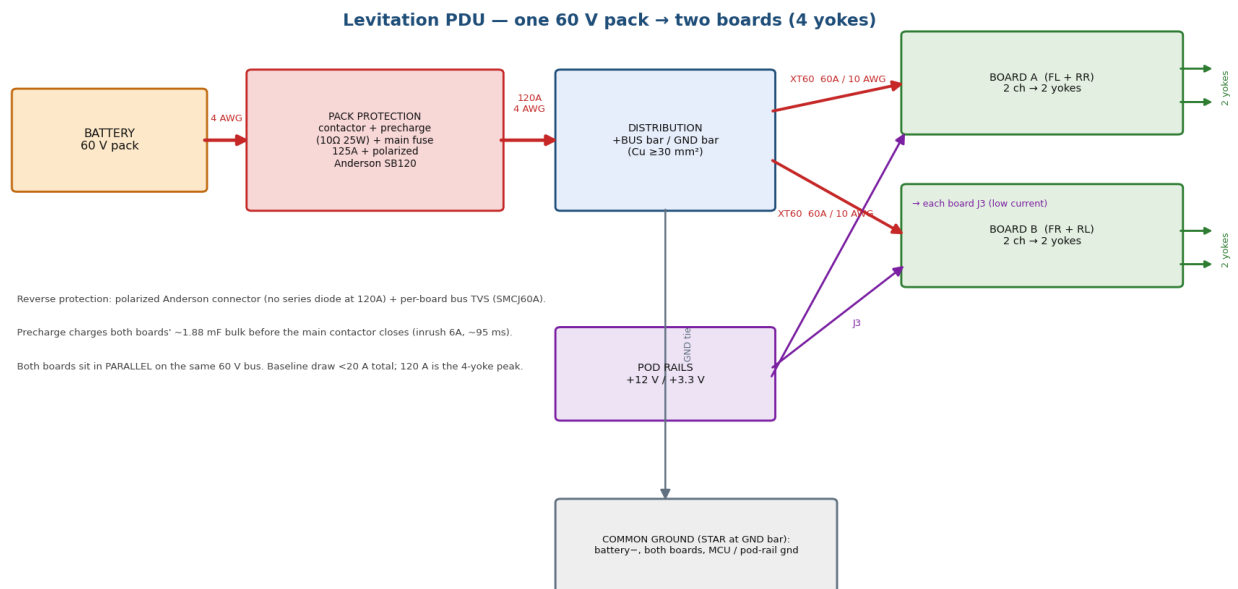
Why a separate PDU instead of wiring the battery straight to each board — it puts all the pack-level protection (main fuse, bus TVS, precharge resistor, and contactor) in one place, sized once and serviceable without touching the drive electronics. Feeding the boards in parallel, rather than daisy-chaining power through one board to the next, means no board ever carries another board's current and every yoke sees the same stiff bus reference.

Why two identical two-channel boards instead of one large four-channel board — modularity and fault tolerance. The smaller boards are easier to fabricate and hand-assemble, they spread the heat

across two heatsinks instead of concentrating it, and because they are interchangeable a single spare covers either position. Above all, splitting the drive means a fault on one board removes only two of the four yokes rather than the whole pod.

Why the yoke pairing is diagonal — Board A drives front-left + rear-right and Board B drives front-right + rear-left, so if a board fails the pod loses two diagonally opposite corners rather than two on the same side or end. A lost diagonal keeps the remaining support roughly balanced about the pod's center, which is far more recoverable than dropping two adjacent corners and pitching or rolling the pod hard to one side.

Why control stays centralized — the MCU/levitation controller and its 12 V and 3.3 V logic rails come from the pod and fan out to both boards. Keeping all four yokes under one controller with a common ground reference is what lets the levitation loop coordinate the four corners together; the boards are power stages that execute the controller's PWM, not independent decision-makers.

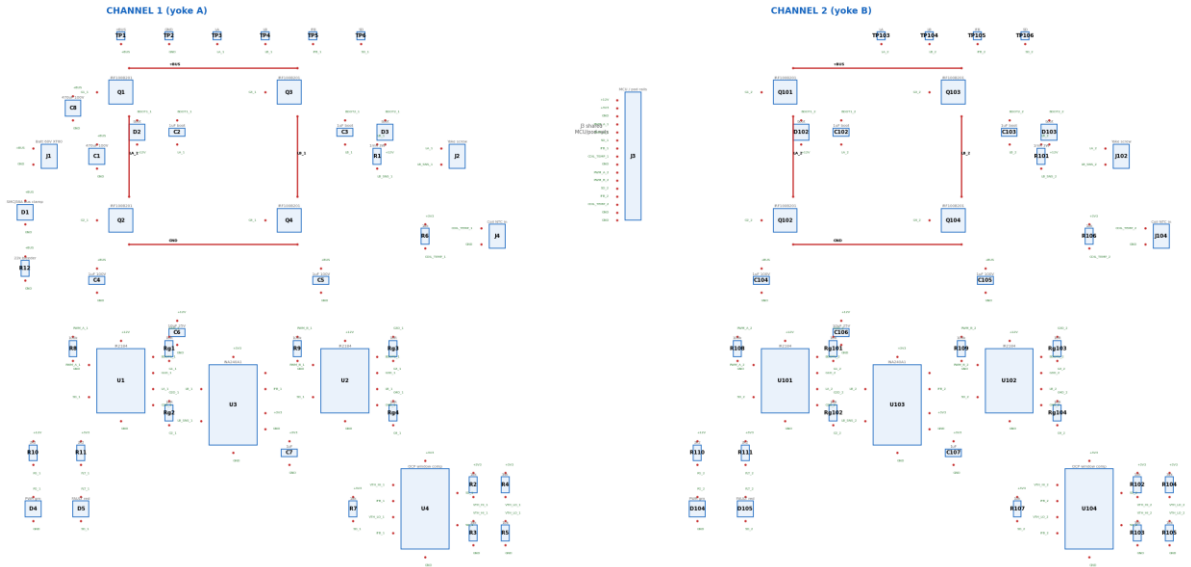


System power architecture: one pack → PDU → two boards → four yokes.

4. The control board (2-channel H-bridge)

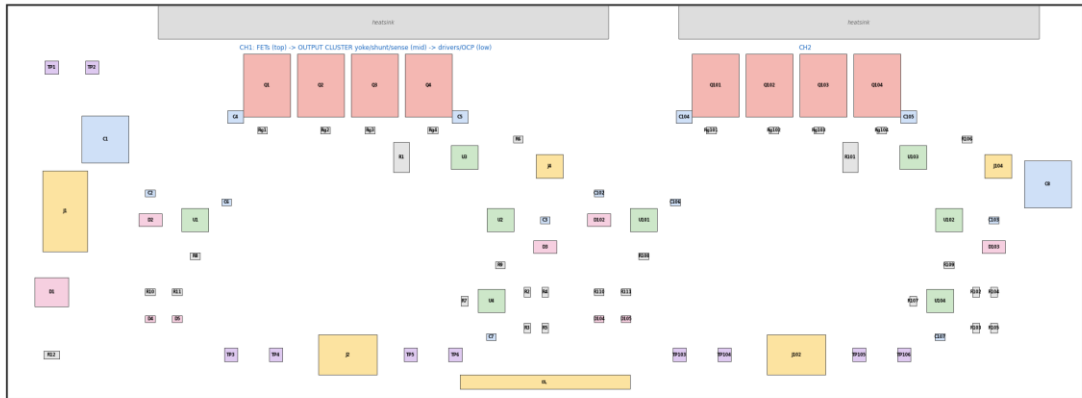
Each board is a two-channel power stage. Per channel: four IRF100B201 power MOSFETs (100 V, 3.5 mΩ) in an H-bridge, two IR2184 gate drivers, an INA240 current-sense amplifier across a 1 mΩ shunt, and an LM393 over-current comparator. FETs sit in a row along the top edge for a bolt-on heatsink; the switch-node copper runs down the middle to the yoke terminal; drivers and logic flank it.

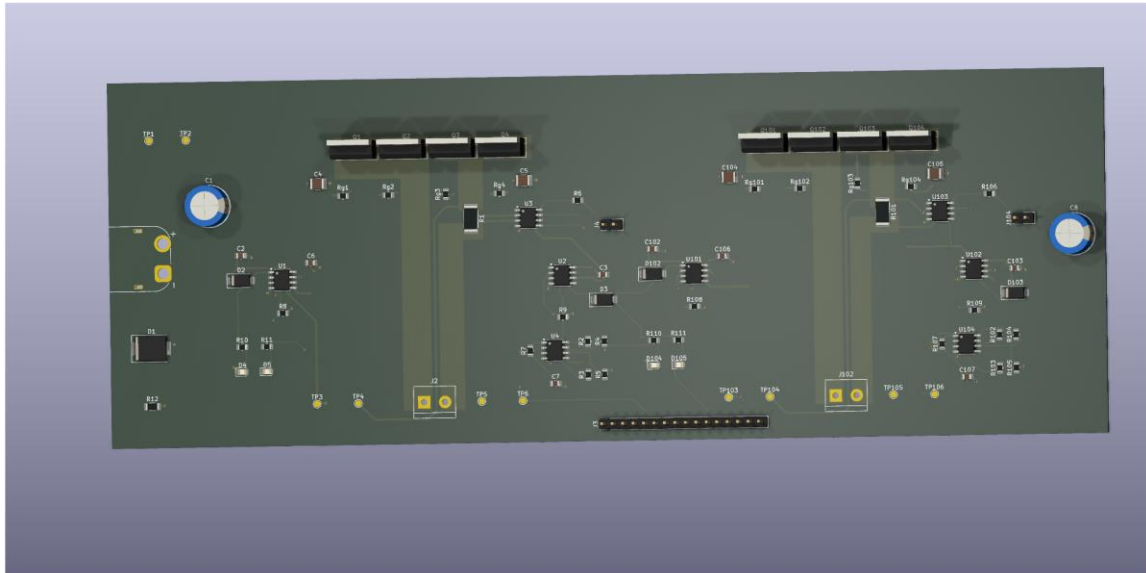
KiCad schematic preview (my render) - FULL 2-channel board (rev 0.9, 86 parts)



Control-board schematic

Placement v3 (yoke/shunt below bridge) 240x88mm, 86 parts



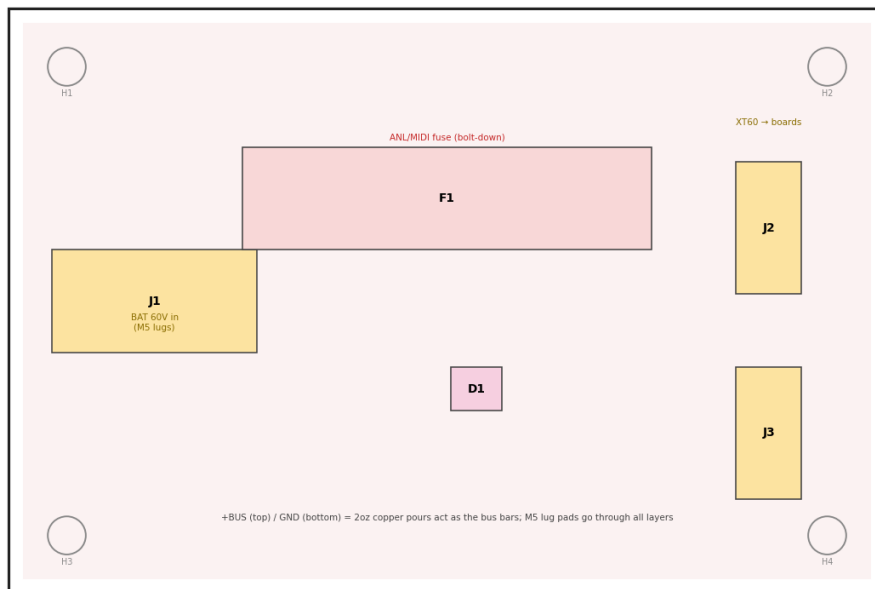


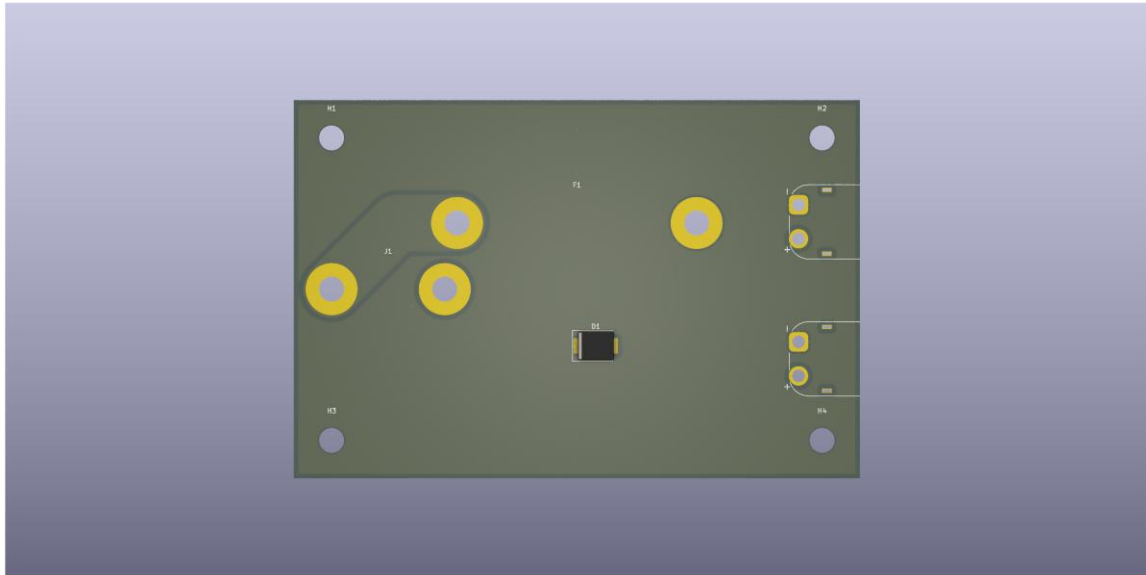
Control-board PCB placement (232×84 mm, 4-layer): FET rows top, output cluster mid, drivers/OCP flanks.

5. The power-distribution unit (PDU)

The PDU takes the battery in through a main fuse and bus TVS and splits it to two XT60 outputs, one per board. It is a simple 2-layer heavy-copper board. Reverse protection is the polarized battery connector; the precharge resistor and main contactor live at the pack. Rated for the 120 A worst case (all four yokes at 30 A).

Levitron PDU — PCB placement (120x80mm, heavy copper)





PDU board: battery lugs → fuse → TVS → two XT60 outputs, heavy copper pours.

6. Validation status

The design has passed every pre-fabrication check we can run on paper:

Check	Result
Schematic electrical rules (ERC)	Clean — 0 errors
PCB design rules (DRC)	Clean (only the expected edge-connector overhang)
Wiring / topology audit	45 / 45 checks pass — every net matches intent
Electrical simulation (numerical + LTSpice)	Current control, ripple 0.6%, freewheel, OCP all correct
Thermal + current-density	Pass with the FET heatsink fitted

Full detail is in the separate Design Validation Report. *Note: these checks verify the design; the physical board is still to be built and bench-tested.*

7. Key parameters & assumptions

Parameter	Value	Note
Bus / battery voltage	60 V	Confirm max charged pack voltage
PWM frequency	16–32 kHz	16 kHz motor-controller floor
Coil per yoke	2 Ω / 5 mH	2× (4 Ω / 10 mH) coils in parallel — measure to confirm
Baseline current	< 5 A / yoke	ideal < 1 A
Peak current	30 A / yoke (120 A total)	transient; confirm if it can be sustained
Over-current trip	45 A	hardware, per channel
Topology	full H-bridge, sign-magnitude PWM	bidirectional coil current
Boards	2 identical, 2 yokes each	diagonal pairing FL+RR / FR+RL
FET heatsink	required, ≤ 7.6 °C/W	mandatory for peak current

8. Bill of materials (per control board)

Qty	Part	Package	Function
8	IRF100B201	TO-220	Power MOSFET (100 V, 3.5 mΩ)
4	IR2184	SOIC-8	Half-bridge gate driver
2	INA240A1	SOIC-8	Current-sense amplifier
2	LM393	SOIC-8	Over-current comparator
2	1 mΩ 3 W shunt	2512	Current sense
2	470 μF 100 V	radial	Bus bulk capacitor
4	1 μF 100 V	1210	Bridge decoupling
1	SMCJ60A	SMC	Bus TVS clamp
~30	R / C / diodes / LEDs	0805/SMA	Gate R, boot, decoupling, thresholds, status
1 / 2 / 1	XT60 / screw term / 16-pin hdr	—	Battery in / yoke out / MCU+rails

PDU adds: battery lugs (or Anderson SB120), a 125 A ANL/MIDI fuse, an SMCJ60A TVS, 2× XT60, and (at the pack) a 10 Ω precharge resistor + main contactor.

9. Open items & next steps

- ☐ Confirm with the lev team: max charged pack voltage, final series-vs-parallel coil arrangement, measured L/R, and whether 30 A can be sustained.
- ☐ Cross-check every part's footprint pin-out against its datasheet (the one error class the audits don't catch).
- ☐ Fabrication review (DFM / Gerbers), then order boards.
- ☐ Staged bench bring-up on the first article: low-voltage switching check on the test points, verify the over-current trip, then ramp to full bus.
- ☐ Controls team closes the current loop / levitation stability (outside these boards).